Spas of distinction.

PACIFIC MARQUIS®

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INDEPENDENCE, OR 97351
(503) 838-0888

Spas of distinction.

PACIFIC MARQUIS®

SPA OWNER'S MANUAL

SUPPORT SYSTEM FOR CUSTOM THERAPY SERIES AND INDOOR/ OUTDOOR SERIES PORTABLE SPAS NO. PM 1000

THIS IS AN IMPORTANT SAFETY DOCUMENT.

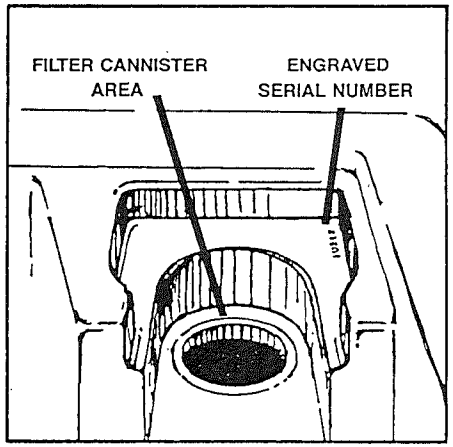
- Keep it in a safe place.
- Pass it on to any new owner.
- If lost, contact your local Pacific Marquis dealer or write to Pacific Marquis and new copy will be provided at no-charge.

SAVE THESE INSTRUCTIONS

Please take a few minutes to read this manual. A thorough knowledge of the basic safety precautions and proper maintenance procedures will provide years of safe and pleasant enjoyment of your Pacific Marquis Spa.

(It should also be noted that in some cases, failure to follow these instructions may result in damage that will not be covered by your warranty or may void the warranty entirely.)

SERIAL NUMBER LOCATION ▶
On Pacific Marquis spas, the serial number is engraved directly above the filter canister area (see diagram). On those models which are listed by Underwriters Laboratories (U.L.), you will also find the serial number engraved on the foil tag attached in the equipment bay. ▶



Please fill out for future reference.

PACIFIC MARQUIS SPA MODEL _____

SERIAL NUMBER _____

CAPACITY (GALLONS) _____

MEMBER

NATIONAL SPA & POOL INSTITUTE

Spas of distinction. PACIFIC MARQUIS®

This symbol on the product's nameplate means it is Listed by UNDERWRITERS LABORATORIES INC.

This spa is designed and intended for residential use only and should not be used in a commercial, public or semi-public installation.

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IMPORTANT SAFETY INSTRUCTIONS

When installing and using this electrical equipment, basic safety precautions should always be followed including the following:

1. READ AND FOLLOW ALL INSTRUCTIONS.

2. DANGER – RISK OF ELECTRICAL SHOCK! Install at least five feet from all metal surfaces.

3. DANGER – Risk of electrical shock. Do not permit any electrical appliance, such as a light, telephone, radio, or T.V. within 5′ (1.52 M) of a spa or hot tub.

4. DANGER – To reduce the risk of injury, do not permit children to use this product unless they are closely supervised at all times.

WARNING – RISK OF CHILD DROWNING. Extreme caution must be exercised to prevent unauthorized access by children. To avoid accidents, ensure that children cannot use a spa or hot tub unless they are supervised at all times.

5. DANGER – To reduce the risk of injury, do not remove suction cover or grate.

6. TO REDUCE THE RISK OF INJURY: The following spa use rules are most important. Using your spa properly and safely will give you countless hours of enjoyment and pleasure. But, just as with many products today, misuse can be extremely harmful. Protect yourself, your family and your guests by following the safety rules listed below:

A. Do not drink alcoholic beverages before or during spa use. One of the effects of alcohol is slowed reflexes and it can make

people drowsy. Jetted hot water also helps induce muscle relaxation. The two together could induce sleep and this, in turn, could possibly lead to drowning. Because of these facts, it is vitally important that you do NOT allow the mixing of alcoholic beverages with a spa usage.

B. Pregnant women and people under medical care (for such problems as heart disease, diabetes, blood pressure, circulatory problems, or obesity) must consult their physician prior to using the spa. Some medications can induce drowsiness. Never use the spa when taking anticoagulants, antihistimines, vasoconstrictors, vasodilators, stimulants, hypnotics, narcotics or tranquilizers. In all cases, if you are not thoroughly familiar with the medication you are taking, check with your doctor prior to using the spa.

C. Spa water temperatures should be maintained in the 98°F to 104°F range. An accurate underwater thermometer is a "must." 100° water is generally safe and enjoyable for most adults. However, remaining in 100° water for an extended period will raise your body temperature to this level and you may experience some discomfort. As a general rule, when you raise the temperature of the spa water, you should reduce the amount of time you soak. Twenty minutes is about the limit for 102°F water; limit your soaking to 10 minutes above 102°F. Never soak in water hotter than 104°F because temperatures above this level may raise your body temperature to a level that could cause drowsiness, fainting, hyperthermia or heat stroke. (Symptoms of hyperthermia are: muscle incoordination and cramps, nausea, rapid heartbeat, pale skin and delirium.) Before entering a spa or hot tub, the user should measure the water temperature with an accurate thermometer since the tolerance of water temperature-regulating devices may vary as much as ±3°C (5°F). It is always important to check the in-spa thermometer before using your spa and to limit your time in the spa as suggested above. Use 98°F to 99°F – normal body temperature – for extended periods of soaking.

A WORD ABOUT PREGNANT WOMEN

Soaking for extended periods in water 102° or higher may effect body temperature to the point that it could affect the fetus. This is especially important during the first 90 days of pregnancy. Spa use at 102° should be limited to no more than 15 minutes. It is strongly suggested that women who are pregnant (or those who are considering becoming pregnant in the near future) check with their doctor on his recommendation for safe spa use.

A WORD ABOUT CHILDREN

As previously explained, soaking in hot water will affect the body's temperature. In small children, the body temperature can increase more quickly than with a full-grown adult. Thus, children are unable to absorb the same amount of heat. Therefore, children's time in the spa should be less than an adult's and children using a spa must be constantly supervised. Children should never be allowed to swim underwater.

D. People with infections, skin sores or open wounds should not use the spa. Warm or hot water may serve as a "breeding ground" for some types of bacteria. It is vitally important to keep your spa properly disinfected with chlorine or other equally effective disinfectants.

E. It is recommended that people shower before and after using the spa. Showering before removes deodorants, lotions, perspiration and oils which may clog the filter. Showering after soaking will help remove any bacteria that may be in a spa that is not fully disinfected.

F. NEVER allow children to use the spa without adult supervision.

G. Adults should not use the spa without someone nearby who can be called should anything unexpected occur. As an added precaution, you may want to post emergency phone numbers near your phone so that they are handy should a need arise.

H. Always use unbreakable containers around the spa; never use glass.

I. Always use caution when entering and exiting the spa. Wet surfaces can be slippery and may cause a fall. Hand rails and non-skid surfaces are recommended.

J. Suction through drains and skimmers can be powerful. Any drain or skimmer with a damaged cover can be dangerous, especially to small children or adults with long hair. Should any part of the body become drawn to a drain or skimmer, turn off the spa immediately. Long hair should be restrained in a bathing cap; never allow long hair to float in the spa. Always replace broken drain outlet cover immediately.

K. Keep electrical appliances, including telephones, away from the spa to avoid a possible shock hazard. Do not try to adjust or touch the spa equipment such as the pump, heater or any other electrical appliance while you are in the spa.

L. Never walk, climb, play or jump on the Energy Cover of your spa. Never swim or play under this cover when it is installed on the spa. Also, do NOT rely on your Energy Cover as a safety cover for children. It is a precautionary measure only and children must be supervised when they are in or around the spa.

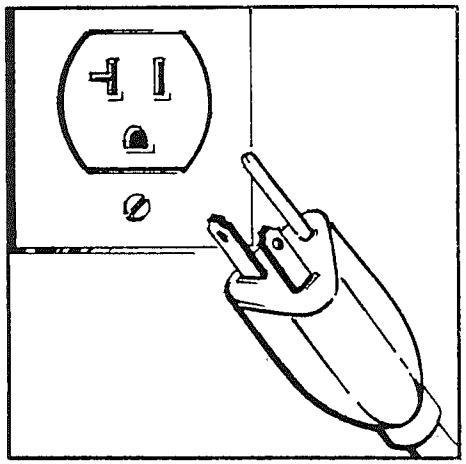
M. A fence around your spa with a self-closing and self-latching gate can be the best protection against unauthorized entry and use. If your spa is indoors, lock the door to the room to keep out unauthorized users.

N. Do not use a spa in conjunction with strenuous exercise without consulting your physician and obtaining his approval.

7. Turn OFF all electric power at the main circuit breaker or disconnect panel prior to performing any service to the spa equipment.

8. It is the responsibility of the spa owner to ensure that all electrical connections are made in accordance with the National Electric Code and any state or local electrical codes in effect at the time of installation. For all permanently connected units the electrical supply for this product must include a suitably rated switch or circuit breaker to open all ungrounded supply conductors to comply with Section 422-20 of the National Electrical Code, ANSI/NFPA 70-1987. The disconnecting means must be readily accessible to the tub occupant but installed at least 5 feet (1.5 m) from tub water.

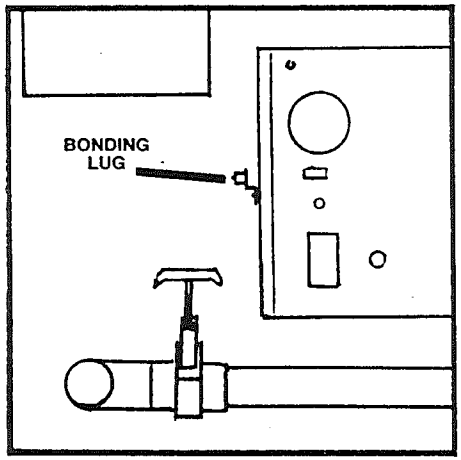
9. Connect 115V support systems only to a grounded grounding type receptacle. DO NOT, under any circumstances, modify the power cord in any way to fit other than a 20 amp receptacle.



DO NOT BURY THE POWER CORD.

WARNING: *To reduce the risk of electric shock, replace damaged cord immediately.*

10. A pressure wire connector is provided on the bottom surface of the control box inside the spa to permit connection of a bonding wire between this point and any metal equipment, metal water pipe or conduit within five feet of the spa as needed to comply with local requirements. The bonding

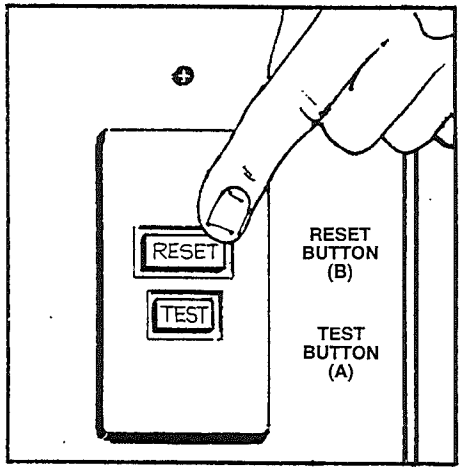


wire must be at least a #8 AWG (8.4 mm^2) solid copper wire. This is to reduce the risk of electric shock.

11. Connections should be made with copper conductors only. All conductors, circuit breakers and/or fuses must be sized in accordance to the Total Amperage Load as specified on the electrical control box data label.

12. All 230V support systems require a three wire plus ground electrical service (line 1, line 2, neutral, ground). All 230V connections can be made by removing the front of the electrical control box.

13. The Pacific Marquis support system is provided with a ground fault circuit interruptor (GFCI) located on the front of the electrical control box. The GFCI is a safety device which protects users from exposure to electric shock hazard.



The GFCI must be tested before **each use** of your spa in the following manner:

(A) With the equipment connected to the proper electrical power source and with the circuit breakers on, push the test button: the reset button will pop outward and the equipment will stop operating.

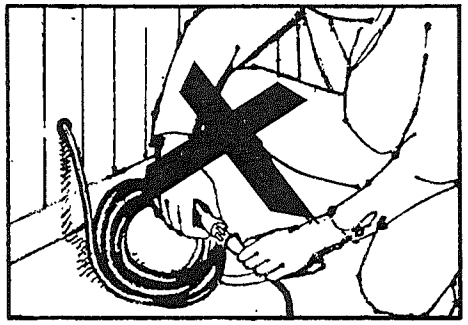
(B) Push the reset button: the reset button should return to its original position.

CAUTION: If the GFCI fails to operate in this manner, disconnect the power cord from the wall receptacle or turn off the circuit breaker at the main electrical panel or disconnect. Failure to operate in this manner means a ground current is flowing indicating the possibility of electric shock. Do NOT reconnect the power until the source of the ground current has been identified and corrected by a qualified service technician.

14. The spa must be installed to provide proper drainage and to keep water out of the electrical equipment components.

The spa must be installed in such a manner as to provide access for servicing all electrical and plumbing components from above or below any decks or floors.

15. Do not use an extension cord. The spa must be placed near enough to the power source to prevent the need for an extension cord.



16. NEVER operate the spa support system when the spa is empty. This could result in severe damage to the heater, pump and spa support components and is a potential fire risk.

CHEMICAL SAFETY

17. Add only one chemical at a time to the spa water. Never mix chemicals or chemical solutions together. Always follow the manufacturer's instructions for dilutions and handling precautions. When diluting, add chemicals to water – not water to chemicals. Avoid skin contact when working with acid. ALWAYS STORE CHEMICALS OUT OF THE REACH OF CHILDREN **(#37).**

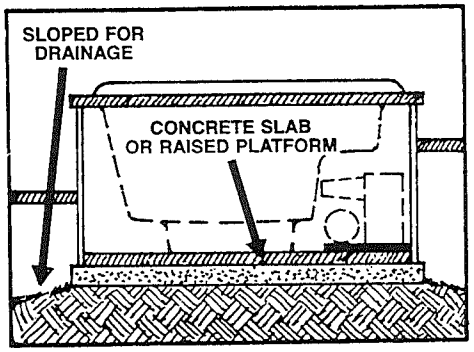
18. SAVE THESE INSTRUCTIONS

INSTALLATION AND USE

LOCATION

19. For your portable spa to function properly and safely, it must be located on a hard, flat, level surface. Inground spas require special installation techniques and should be planned in conjunction with your Pacific Marquis dealer. Improper installation can result in structural damage to the spa and a voiding of your spa warranty. Items to discuss with your dealer include local construction codes, electrical service requirements, accessibility and serviceability of equipment and components, present underground obstructions such as gas, water and telephone lines, safety measures such as fences and locks and visibility of the spa installation from the home, street and neighbors.

You must allow for sufficient water drainage around the spa to help preserve the wood skirt and support structure and to provide adequate drainage of the equipment bay for the components.



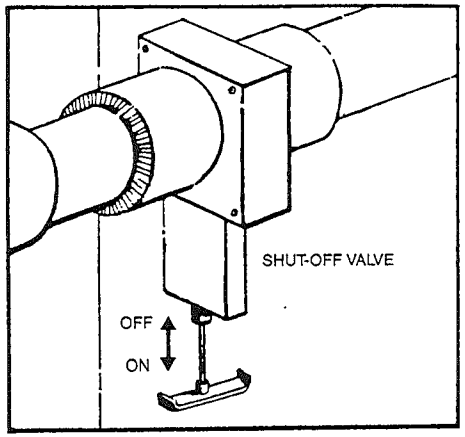
Your spa MUST be installed to permit access for servicing the equipment above and/or below any decks or floors. Access is essential and must allow adequate room for service personnel.

Be sure the spa is in the final position BEFORE filling with water. Check for power cords, tools or hoses which may be caught underneath the spa.

CAUTION: *Never try to move a spa that has not been fully drained. To do so can result in damage to the spa and physical injury to the mover.*

SHUT-OFF VALVES

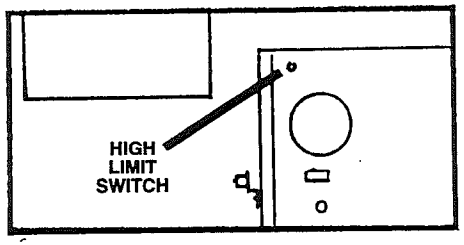
20. Your spa is equipped with two SHUT-OFF VALVES that can completely shut off the water flow to the support system for dealer servicing. At times, a new spa or one that has recently been serviced may have the SHUT-OFF VALVES partially closed which can restrict the water flow and hinder jet performance. Be sure the valves are ALWAYS fully open with the handles fully extended



HEATER HIGH-LIMIT SWITCH

21. The high-limit switch will stop operation if the temperature within the heater assembly reaches a factory-set non-adjustable limit.

Several conditions could cause the switch to activate: low water level, shut-off valves closed while heater is on, dirty filter, restricted or clogged plumbing lines.

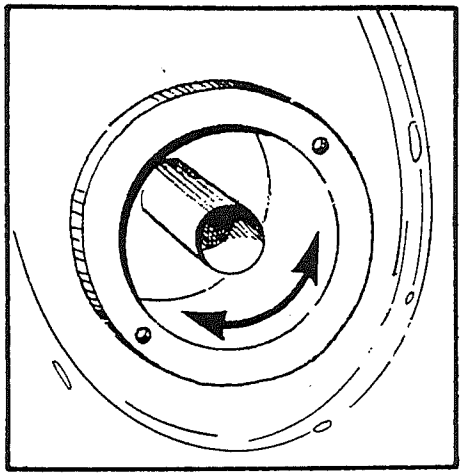


22. To reset the high-limit switch, push the red button located on the upper left hand corner of the control box panel.

The switch cannot be reset until water temperature within the heater assembly drops to a lower, preset temperature. If the switch trips repeatedly, the support system should not be operated until the problem has been corrected. Please refer to your Pacific Marquis Dealer.

HYDROJETS

23. Your spa is fitted with directional hydrojects. The jet action can be directed 360° by aiming the swivel ball nozzle.



SPA SIDE CONTROLS

24. The SPA SIDE CONTROL places all of your spa's functions within arm's reach.

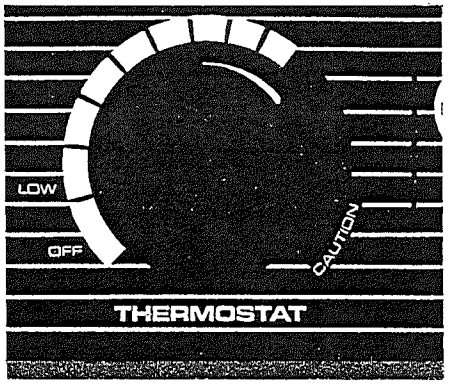
PUMP

25. The button marked "PUMP" controls the two speed pump. Pushing the button either activates the HIGH speed mode for maximum jet action or returns the pump to LOW speed for circulation.

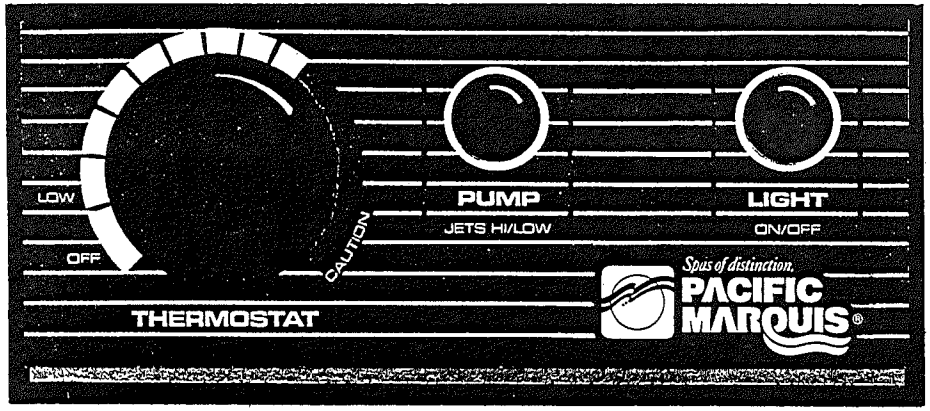
CAUTION: Do not leave pump on high speed when spa is not in use. Overheating of the spa water can result.

HEATER THERMOSTAT

26. The thermostat for the heater is located on the left hand side of the spa side panel. To adjust the spa's heat, turn the thermostat knob clockwise to increase the temperature and counter-clockwise to decrease the temperature.



The length of time it takes the water to reach a desired temperature depends on several factors. Raw water temperature, ambient air temperature, spa size, relative humidity, type and insulative qualities of a cover and constancy of electric power are all factors.



DO NOT EXPECT TO FEEL HOT WATER FROM THE JETS AT INITIAL START-UP.

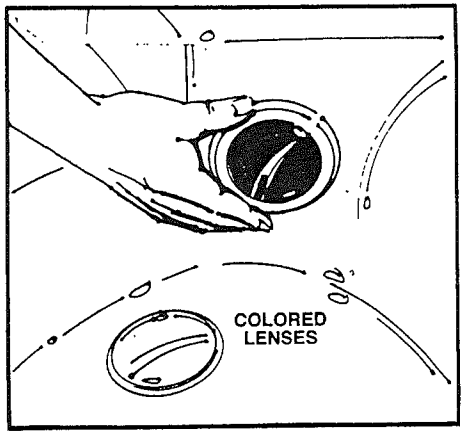
CAUTION: The spa should never be operated at temperatures above 104°F. An accurate underwater thermometer is vital in monitoring water temperatures. If the thermostat allows the heater to heat the spa to temperatures greater than 104°F, contact your Pacific Marquis dealer. Refer to Safety Precaution "C" in the front of this manual for further information about recommended water temperatures.

UNDERWATER SPA LIGHT

27. Your spa is equipped with an underwater spa light located in the spa footwell. The low-voltage system produces a bright white light for evening use in the spa and makes entering/exiting the spa safer.

The light control switch is located on the Spa Side Control panel.

You may install one of two interchangeable colored lenses over the white lense housing. The lenses snap on and off the housing and can be installed even when the spa is full.

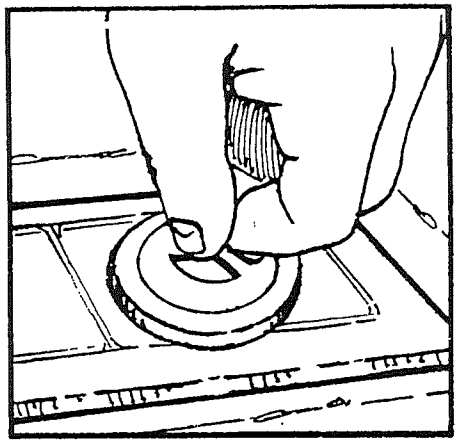


AIR CONTROL VALVE

28. Your spa is equipped with a single air control valve which allows regulation of the water action from the jets.

The water action can be increased by turning the air control valve ¼ turn in either direction until 'ON' can be seen through the view window.

NOTE: Because heat loss occurs as air is injected into the water, close the air control valve completely when the spa is not in use.



TIME CLOCK OPERATION (General)

29. Your spa is equipped with a single 7-day time clock built into the electrical control box. The time clock is designed to give you maximum operating convenience and economy. By using this time clock the water can be filtered daily or, several times a day, to maintain sparkling clear water with ideal chemical distribution.

The timeclock is provided as a convenience. It is still best to operate your spa 24 hours a day in the low-speed circulation mode (when not in use). This will provide the maximum amount of filtration as well as prolonging the life of the pump and other electrical components.

ADJUSTING THE TIME CLOCK

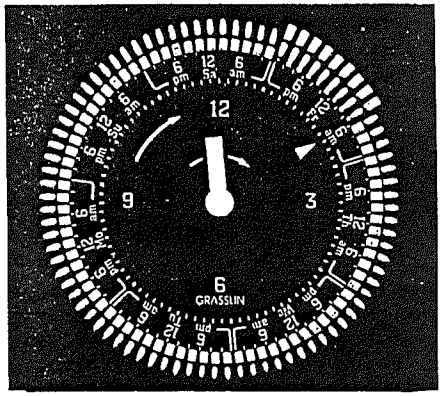
NOTE: Any time the circuit breakers are shut off or the spa power cord is removed from the outlet, the clock will need to be reset to the actual time of day.

TIMECLOCK
(7-Day Operation)

30. CLOCK SET POINT: Rotate outer clock ring clockwise and set correct time of day opposite Set Point arrow.

TIME CLOCK LUGS: (Each lug represents a 2-hour period of a selected day of the week.) Push outward from center of clock to set ON times. The example shows ON at 6:00 a.m. Friday and OFF at Midnight Friday. To cancel settings, push inward on the lugs.
If the thermostat is calling for heat and the time clock reaches an ON time, the PUMP will start and the HEATER will turn ON. When the water reaches the desired temperature, the thermostat will turn the HEATER OFF and the pump will continue. This cycle will continue during each ON period as determined by the setting of the time clock lugs. Wait a few days and adjust filtration times as necessary to maintain sparkling clear water with a minimum of ON times.

HEATING/FILTERING TIMER



TIME CLOCK BYPASS SWITCH

31. In the TIMER CONTROL position, the time clock has control and will start and stop the filtration cycle as determined by the settings of the TIME CLOCK lugs (#30).

In the CONTINUOUS OPERATION position, the filtration cycle will be continuous. This is desired following periods of heavy usage or when adding chemicals.

The time clock setting is not disturbed and timed cycles will begin again when this switch is returned to the TIMER CONTROL position.

NOTE: If your spa is equipped with an ozone sanitation system, refer to the supplemental booklet dealing with the ozone system for complete instructions and chemical guidelines.

WATER CHEMISTRY GUIDELINES

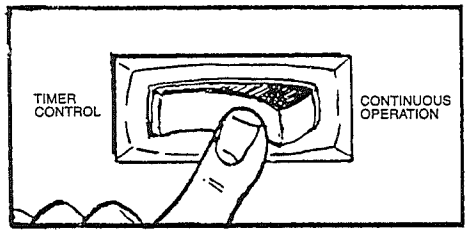
32. When filling the spa with water, use the chemicals recommended by your PM Dealer to bring the water to a correct chemical balance (#32-36). **NEGLECT IN MAINTAINING THE CORRECT WATER CHEMISTRY CAN VOID YOUR SPA WARRANTY!**

There are several methods available to sanitize the water in your spa. Pacific Marquis recommends using chlorine, bromine, or ozone. No other oxidizing agents are suggested. Use of other oxidizing agents may damage equipment and void the warranty.

33. CAUTION: It is essential that you shock chlorinate your spa after filling it with water BEFORE you begin to use it. Directions for shock or super chlorination are given on the chlorine concentrate container. This shock treatment will give an initial sterilization to your water, kill any algae spores that may be present and oxidize undesirable minerals and organic matter in the raw water.

This treatment must be repeated anytime the spa is refilled or the water has lost its chemical control due to neglect or lack of use.

It is recommended that sodium dichlorocyanurate (DyCloro) or lithium hypochlorite be used for spa use. Calcium hypochlorite is a type of chlorine which is used in swimming pools and it is not recommended for spa usage.



CHEMICAL LEVELS

34. pH levels should be kept between 7.2 and 7.6. Failure to maintain these levels can result in burning of the eyes, skin rashes and other discomforts as well as clogged pipes, staining and equipment damage.

35. Alkalinity should be maintained between 100-150 parts per million (ppm).

36. Chlorine levels should not exceed 3 ppm except during periods of super chlorination **(#33).** Super chlorination is used only to treat raw water after completely draining the spa, and after periods of heavy use.

Since water chemistry and hardness varies from region to region, you should contact your Pacific Marquis dealer for advice on how to best treat your water. He has a complete line of chemicals to help you get optimum use and enjoyment from your spa.

CHEMICAL SAFETY

37. It is important that you read and follow the safety tips for chemical usage and storage listed here. This valuable information is contained on pages 11 and 12 of the "Sensible Way to Enjoy Your Spa or Hot Tub" issued by the *National Spa and Pool Institute* (NSPI) who have granted us permission to use this information here.

The chemicals needed for your spa or hot tub help make it clean, disinfected and more attractive to use. But remember that these chemicals are potentially dangerous and may present some hazards if not used properly. Carefully follow the manufacturer's instructions for the use and storage of chemicals.

In general, here are some tips for chemical use and storage:

- Before using chemicals, read the labels and directions carefully. Follow label use instructions.
- Keep all chemicals out of the reach of children.

STORAGE

- Chemicals for test kits should be replaced each year.
- Keep the original lids on all chemical containers and make sure the lids are closed tightly when not in use.
- Do not stack different chemicals on top of one another.
- Store your spa or hot tub chemicals in a clean, cool, dry, well-ventilated area, preferably off the floor, to prevent contamination from other materials. Keep them away from chemicals and equipment used for garden and lawn maintenance.
- Keep liquid chemicals away from dry chemicals. Keep apart chemicals which are different forms of oxidizing compounds. Physically separate all different forms of chemicals.
- Do not store your spa or hot tub chemicals where other flammable items may mix with them. For example, a mixture of these chemicals and fertilizer can cause a fire or explosion.

USAGE

- Never mix two chemicals together. Use a clean scoop for each chemical, and avoid combining materials from "old" and "new" containers.
- Test the water in your spa or hot tub with a reliable test kit on a schedule recommended by your spa or hot tub professional. Add the necessary chemicals according to the test results and the manufacturer's instructions. In the hot water environments of spas and hot tubs, disinfectants may rapidly break up and spread out, requiring more frequent water testing. Follow your manufacturer's instructions in this regard. The more people who use the facility, the more frequently you should test the water.
- Do not inhale dust or fumes from any chemicals. If necessary, use proper protective devices for breathing, handling and eye protection. Promptly wash off any residues which get on your skin.

- Never reuse old containers unless specified by the manufacturer.
- If you have any questions regarding safe handling, storage or use of spa or hot tub chemicals, contact the manufacturers.
- Always add the chemicals directly to the spa or hot tub water, either in a suitable feeder, or distributed across the surface of the water or diluted and poured into the water. Follow label use instructions.
- When preparing water solutions for feeder application, pour the chemical slowly into the appropriate amount of water, stirring constantly to provide mixing and dilutions.
- Always add chemicals to water. Never add water to chemicals.
- Never add chemicals to the spa or hot tub water while people are using the facility.
- Carefully clean up any spilled chemicals with large amounts of water, to dilute and wash away the chemicals. Disinfectants and pH adjustment chemicals can usually be sent to the sewer with large quantities of water, since they are intended for use at low levels.
- Wash out empty, disinfectant containers before disposing to eliminate danger of fire, explosion or poisoning.

ELECTRICAL REQUIREMENTS

38. Before start-up or performing any service to the spa equipment, turn OFF all electric power at the main circuit breaker or disconnect panel.

It is the responsibility of the spa owner to ensure that all electrical connections are made by a qualified electrician in accordance with the National Electric Code and any state and local electric codes in effect at the time of installation.

All connections must be made in accordance with the wiring diagram found on the inside of the electrical box cover or the wiring diagrams within this manual.

Connections should be made with copper conductors **only.** All conductors, circuit breakers and/or fuses must be sized in accordance to the Total Amperage Load as specified on the electrical control box data label.

This equipment is designed to operate on 60Hz Alternating Current only at a voltage of 115V or 230V as required.

A pressure wire connector is located on the exterior of the electrical control box. If the installation includes a common bonding grid (reinforced concrete slab, ground plate beneath the spa or any metal water pipe connection), this pressure wire connector should be bonded with at least #8 AWG copper wire to any metal ladder, water pipe or other metal within five feet of the spa.

115 VOLT INSTALLATION

39. Cord connected 115V units must only be connected to a grounded, grounding type 20 amp receptacle. If a 115V unit is to be "hard wired" rather than use the optional 115V cord, it requires a two wire plus ground (line 1, neutral, ground) electrical service. Refer to the diagram on the electrical box cover.

Note: On 115V installations the heater operates only when the pump is on low speed.

230 VOLT INSTALLATION

40. Units to be operated at 230V must have all connections made by a qualified electrician in accordance with the National Electric Code and any state and local electric codes in effect at the time of installation. All 230V units require a three wire plus ground (line 1, line 2, neutral, ground). Refer to the diagram on the electrical box cover.

Verify that the unit had been properly converted to 230V usage as per the instructions on 115V to 230V conversions. **(#41).**

115V to 230V CONVERSION

41. These modifications must be performed by a qualified licensed electrician.

A. Remove the control box cover from the front of the electrical control box.

B. Disconnect the factory supplied 115V cord from the terminal block and remove the cord strain relief grip and 115V cord.

C. All wire changes refer to the LEFT side of the twelve position terminal block. Numbering from the top.

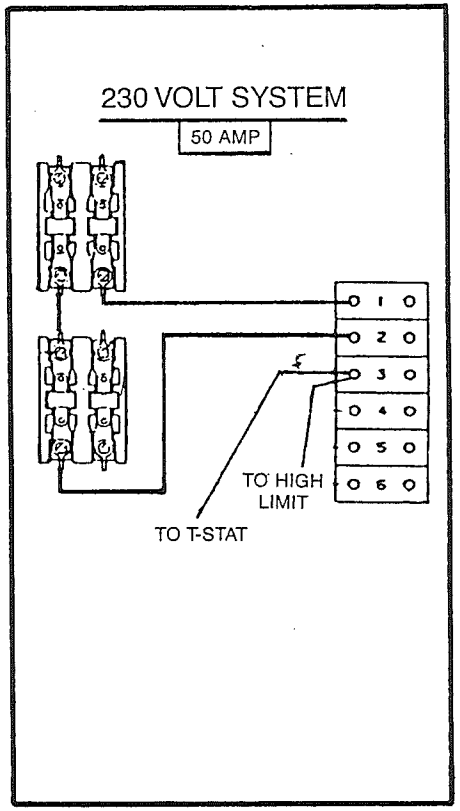
D. Move the black wire from #3 to #1 position.

E. Move the black wire from #4 to #2 position.

F. Move the black wires from #5 and #6 to #3.

NOTE: Do not jump neutral to ground.

NOTE: All 240V field supplied wires must be sized and installed in accordance with any and all national, state and local electrical requirements in effect at the time of installation or conversion.



NOTE: When wired 230V there must be 4 wires.
#1 Hot to contactor
#2 Hot to contactor
#3 Neutral
#4 Ground

STARTING YOUR SPA

42. Make sure power supply is off. On 115V systems the power cord must be unplugged. On 230V systems the circuit breaker in the residence electrical panel must be off.

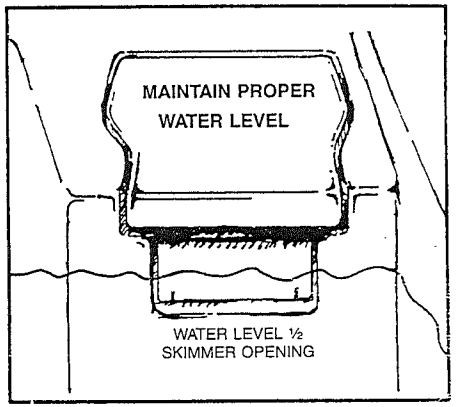
43. Check to see if properly wired for 115V or 230V.

44. Tighten all disconnects, open both shut-off valves **(#20)** and clean all debris from the spa.

Make sure the suction fitting cover is in place in the footwell of your spa. This unit is not to be installed without a suction fitting cover approved to prevent hair and body entrapment. An approved fitting is supplied by the factory with your Pacific Marquis Spa.

45. Select the mode of operation that you desire and move the filtering bypass switch to the desired position **(#31).** Refer to the section on TIME CLOCK for further details **(#29-31).**

46. Fill the spa to the middle of the skimmer opening. **Shock chlorinate** the water before entering and using the spa. It is essential that this shock treatment be done after each refill **(#33).**



NOTE: If the spa is equipped with an optional ozone sanitation system, refer to the supplemental manual for instructions on the sanitation steps needed prior to using the spa.

47. MAKE SURE THERMOSTAT IS IN "LOW" POSITION! **(#26).**

48. Turn electrical power on to the spa. On 115V systems, plug in the power cord **(#11).**

On 230V systems, flip on the circuit breaker in the residence electrical panel.

Note: On 115V installations the heater operates only when the pump is on low speed.

49. Your Pacific Marquis spa is equipped with a door safety system. If the equipment bay door is opened, or ajar, the support system will not operate. THE DOOR MUST BE CLOSED COMPLETELY FOR ANY OF THE SUPPORT SYSTEM COMPONENTS TO BE ACTIVATED.

OPERATING THE SPA

50. Allow the spa to circulate on high speed for 4-5 minutes to discharge air from the plumbing system. After a full flow of water has been obtained from all the jets, adjust the thermostat to the desired setting. Do not expect hot water immediately from the jets. The heater will take several hours to heat the spa to the desired temperature.

NOTE: Your Pacific Marquis spa has been filled and test run at the factory. The first time your spa is filled with water, some discoloration from residual test water may appear. This will disappear when the spa is completely filled and the filtration system is activated.

51. Retest the water for the proper chemical balance and adjust as necessary **(#32-37).** PROPER BALANCE IS IMPORTANT.

For longer pump life, more efficient heating and lowering operating, we recommend the time clock bypass switch (#31) be left on "CONTINUOUS OPERATION." There is less wear on the equipment when it is run at longer periods rather than short intervals.

An unused spa should circulate for at least four (4) hours per day to maintain proper filtration and water temperature. More frequent use of your spa and cooler air temperatures will require longer circulation times.

ENERGY COVER

52. When the spa is not in use, the Pacific Marquis Energy Cover should be kept on the spa to retain the water's heat and to keep out dirt, leaves, etc. Your preset heater thermostat will maintain a constant water temperature between spa uses for a very low cost per day.

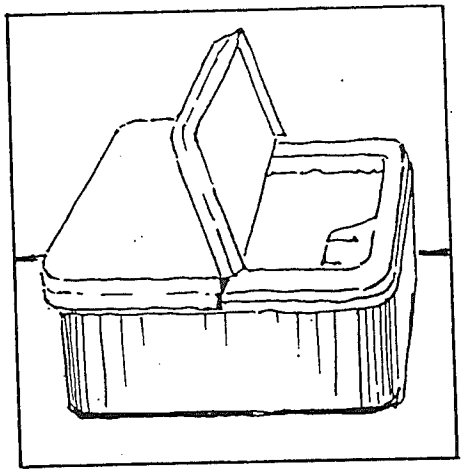
While the Energy Cover is not intended to serve as a safety device, it may serve as a deterent to unauthorized spa use. Cover locks for the Energy Cover are available from your Pacific Marquis dealer and should be installed.

FREEZE PROTECTION

53. Your Pacific Marquis spa has several features which allow you to choose the degree of freeze protection that is most appropriate for your particular spa usage and environment. If the spa is not going to be fully drained and winterized **(#62),** then maximum freeze protection is obtained by the following:

- air switch control on low speed pump **(#25).**
- flip time clock bypass switch to continous operation **(#31).**
- thermostat **(#26)** on the LOW setting or higher

This results in a continuous circulation of the spa water with the heater automatically maintaining a water temperature of approximately 50°F. Regardless of the extent of freeze protection required, always leave the thermostat set on the LOW position or higher and where possible, leave the spa in continuous operation and thermostat control.



REGULAR MAINTENANCE

CEDAR CABINET

54. Your spa cabinet has been treated with a sealant at the factory. It is suggested that the cabinet be treated twice a year with an additional coat of sealant for maximum weather protection. You should contact your Pacific Marquis Dealer for advice on which sealants work best in your environment.

DRAINING THE SPA

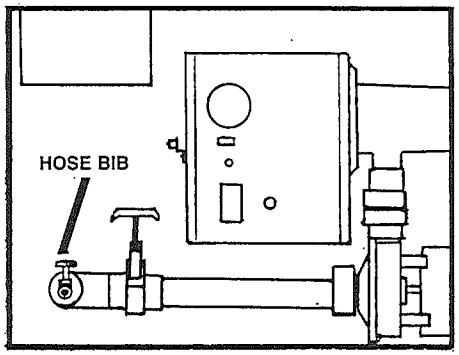
55. It is recommended to completely drain the spa AT LEAST four times a year. More frequent draining may be required depending on use. Unless this is done regularly, the water becomes chemically "saturated" and will no longer respond to regular chemical upkeep. The spa should also be drained before long periods of disuse or for equipment repair. An empty spa should be covered. DIRECT SUNLIGHT ON THE ACRYLIC SURFACE CAN CAUSE SEVERE DAMAGE OR BLEMISHING OF THE SPA SHELL AND CAN RESULT IN THE VOIDING OF ANY SURFACE WARRANTIES.

Because of the small water volumes involved with spas, sometimes it is much simpler and less expensive to drain the water than to adjust the water chemistry with chemical additives.

56. To drain the spa for cleaning or service complete the following steps.

A. Turn the thermostat to the low position **(#26).**

B. Unplug the electrical cord or turn OFF the circuit breakers **(#8-12).**



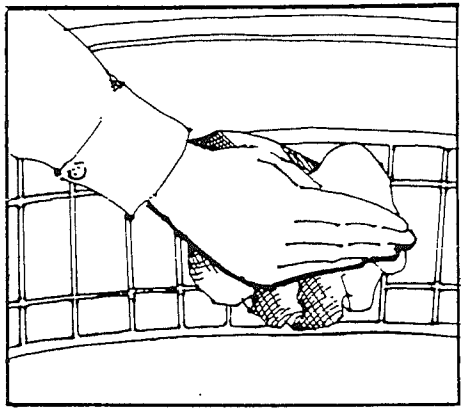
C. Attach a garden hose to the hose bib in the equipment bay.

D. Open the hose bib fully, and the water will drain out by gravity.

E. When the spa is empty, close the hose bib and remove the hose.

CLEANING

57. Should the water line become soiled, it can be cleaned with a soft sponge or cloth. Do not use any abrasive cleaners as they can scratch or dull the spa surface.



58. Your Pacific Marquis Energy Cover can be cleaned with a non-abrasive household cleaner on both top and bottom sides. A good quality NON-SILICONE based vinyl restorer will help protect the surface from the sun's rays.

REMOVING THE FILTER

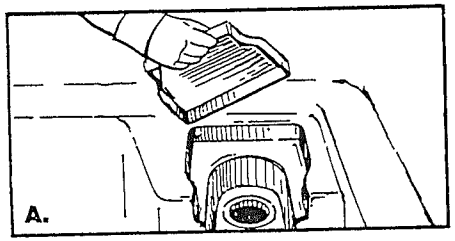
59. Refer to the Spa Maintenance Schedule for cleaning frequency and instructions. The filter cartridge can be removed by completing the following steps:

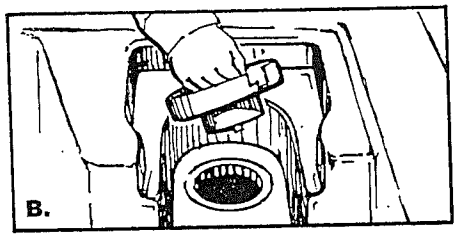
A. Remove the skimmer lid.

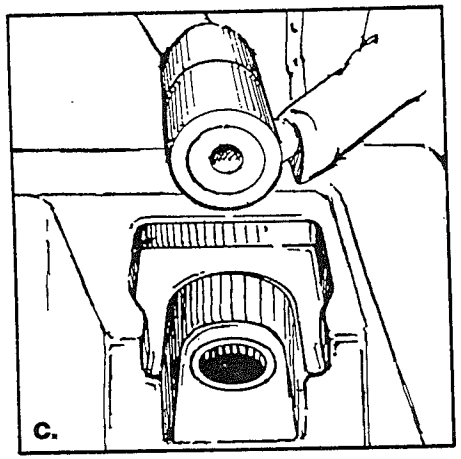
B. Twist the catch basket counterclockwise and lift out.

C. Lift out the filter cartridge. Note the position of the filter: the large hole faces DOWN.

D. To reinstall the filter, simply reverse the procedure outlined above.







REFILLING

60. Fill your spa with fresh water from a garden hose **(#46)** while adding the necessary chemicals **(#32-37).** Remember, it is essential to shock chlorinate after each refill before entering and using your spa. Follow the same procedures as outlined in STARTING YOUR SPA **(#42-49).**

SPA LIGHT BULB REPLACEMENT

61. To replace a burned out bulb, open the equipment door to find the spa light located to the rear of the equipment. The light housing is accessed in this manner.

Remove the lamp socket from the back of the light assembly by turning and pulling at the same time. Gently pull the bulb out of the lamp socket, and replace it with bulb, part #69-0436, available from your Pacific Marquis Dealer. Install the lamp socket back into the light assembly, and close the equipment bay door.

WINTERIZING

62. Your Pacific Marquis spa was designed for year round use and enjoyment and many people find the combination of hot water, hydrotherapy jets and cool air temperatures to be especially soothing. However, if you decide to discontinue the use of your spa for the winter, or for any other extended period of time, we suggest you follow the steps below.

NOTE: Damage can occur to equipment during long periods of inactivity/non use due to condensation.

(See diagram on Page 20 for reference.)

A. Turn thermostat to the LOW position **(#26).**

B. On 115V units, unplug the power cord. On 230V units turn OFF the main circuit breaker or disconnect panel. **(#8-12).**

C. Drain the spa **(#55-56).**

D. Remove all residual water from the seats and footwell. If necessary, use a bucket to bail out the remaining water, then dry the spa with towels. A wet/dry vacuum may also be used.

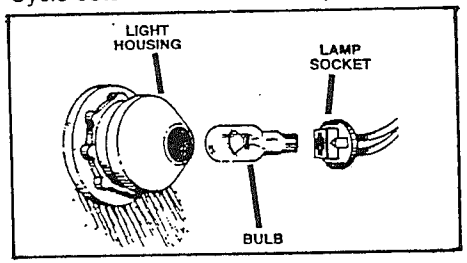
E. Wipe down the tile line **(#57).**

F. Remove, clean, dry and reinstall the filter cartridge **(#59).**

G. Close all air controls.

H. Using the wet/dry vacuum, place nozzle over each jet orifice to remove water from plumbing lines, starting with the highest jet and finishing with the lowest jet.

NOTE: To insure complete draining of plumbing lines, place vacuum nozzle over Cyclo Jets in 'ON' and 'OFF' position.



I. Unscrew and disconnect plumbing lines at heater (13), and both suction (14) and discharge (14) of pump(s).

J. Again, using wet/dry vacuum, place nozzle over the pump inlet (4) to remove excess water.

IMPORTANT NOTICE: For maximum winterizing protection, the pump should be removed from the spa and stored in a climate controlled room.

K. Reinstall the Pacific Marquis Energy Cover (#52). Contact your Pacific Marquis dealer about the availability of a protective spa cover which encloses the entire spa, including the Energy Cover.

SPA MAINTENANCE SCHEDULE

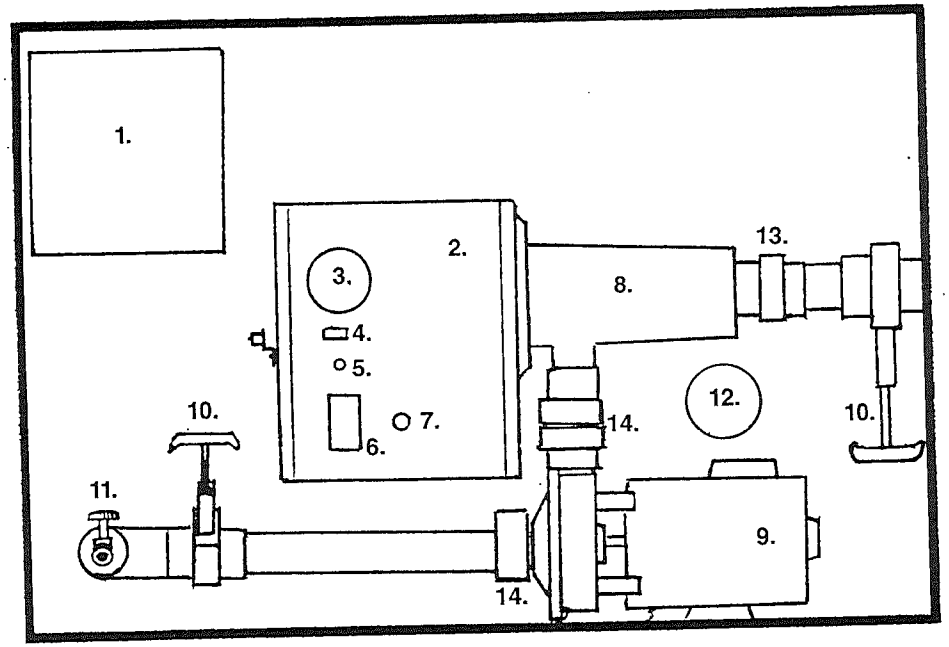
63. DAILY	Replenish the water level by filling up to ½ the skimmer opening **(#46).** Check and clean the catch basket **(#59).** Check the water balance (pH). Adjust if needed **(#32-37).** Check chlorine level and adjust if needed **(#36).** For proper filtration, the water should be circulating for at least four hours a day **(#51).**
64. WEEKLY	Wipe down the water line **(#57).** Remove the filter cartridge **(#59),** and rinse thoroughly with a hose and high pressure nozzle. Soak in a solution of trisodium phosphate (TSP) if the spa is equipped with an ozone sanitation system.
65. MONTHLY	Soak the filter cartridge in a solution of trisodium phosphate (TSP) or any strong, nonsudsing detergent. Rinse thoroughly in clear water. Clean the spa Energy Cover **(#52).** Test the GFCI unit **(#13, 14).**
66. EVERY THREE MONTHS	Drain the spa completely, refill with water and replenish chemicals **(#55,60,32-37).** While the spa is drained, the interior can be sponged with a spa cleaner and rinsed **(#57).** NOTE: Do not wax the surface as the wax will dissolve into the water and clog the filter.
67. EVERY SIX MONTHS	Oil wood skirt with protective sealant.

TROUBLESHOOTING

PLEASE NOTE: The following corrective actions may be performed by the spa owner. If the trouble cannot be corrected in the steps below, please refer to your Pacific Marquis Dealer for service. (Improper servicing by an unauthorized serviceman or spa owner could result in damage not covered by the warranty or could cause serious injury.)

SYMPTOMS	CORRECTIVE ACTION
Equipment does not operate	Make sure the equipment bay door is completely closed. Check to see if the power cord is plugged in (115V operation) **(#8-12).** Check the circuit breaker in the main circuit panel. Test the GFCI to see if it has "tripped" **(#13-14).** IMPORTANT: If the GFCI trips repeatedly, do not use the spa, and contact your Pacific Marquis Dealer.
Pump does not work (in low speed circulation)	Check that the shut-off valves are completely open **(#20).** Flip the HEATING/FILTERING time clock switch to CONTINUOUS OPERATION **(#31).** Check for blockages or restrictions at the main drain and catch basket **(#44,59).** Inspect the filter cartridge for dirt and debris. Clean or replace **(#59).**
Pump does not work (in high speed circulation)	Push the air switch button marked JETS HI/LOW. Check the above items in "Pump does not work (in low speed circulation)."
Inadequate jet action	Push the air switch button marked JETS HI/LOW. Check the spa water level. Water must be ½ the depth of the skimmer opening **(#46).** Check that the shut-off valves are completely open **(#20).** Be sure that the air control valve is open **(#28).** The filter cartridge may be clogged. Pull the cartridge from the canister **(#59).** If the jet action improves, clean or replace the filter **(#59,64).**
No heat	Check all steps above in "Equipment does not operate." Check the High-Limit Control switch in the equipment bay. Push to reset **(#21, 22).** Check that the thermostat dial is set to the desired setting **(#26).** Remember: Do not expect hot water immediately from the jets. The heater will take several hours to heat the spa to the desired temperature. Inspect the filter cartridge for dirt and debris **(#59).** Prolonged use of the jets will have a significant cooling effect on the water. Turn off all blower and jet action to allow the heater to raise the spa temperature. NOTE: On 115V installations the heater operates only when the pump is on low speed.
Underwater spa light does not work	Push the button marked "Light" on the Spa Side Control **(#27).** Tighten the light bulb in the socket **(#61).** Replace the light bulb **(#61).**
Water is cloudy	Check the water chemistry and balance as needed **(#32-37).** Clean or replace the filter cartridge **(#59, 64,65).**

SUPPORT SYSTEM



1. OZONE GENERATOR
2. CONTROL BOX
3. TIME CLOCK
4. TIME CLOCK SWITCH
5. 25 AMP BREAKER
6. G.F.I. GROUND FAULT INTERRUPTER
7. DOOR INTERLOCK
8. HEATER
9. PUMP
10. SLICE (SHUT OFF) VALVE
11. HOSE BIB (DRAIN)
12. LIGHT TUNNEL
13. HEATER CONNECTION
14. PUMP CONNECTION

WARRANTY REGISTRATION

REGISTRATION MUST BE RETURNED WITHIN TEN (10) DAYS OF PURCHASE.

NAME OF PURCHASER _____ DATE PURCHASED _____

ADDRESS _____

CITY _____ STATE _____ ZIP _____

YOUR SIGNATURE _____ TELEPHONE _____

SPA MODEL _____ COLOR _____ SERIAL NUMBERS SPA / EQUIP. _____

DEALER'S NAME _____

DEALER'S ADDRESS _____

RETAIL PURCHASE PRICE $ _____

I HAVE RECEIVED AND READ THE OWNERS MANUAL ☐

I HAVE READ THE SAFETY PRECAUTIONS FOR MY SPA ☐

OPTIONAL INFORMATION:

Age: ☐ 18 to 30, ☐ 31 to 45, ☐ 46 to 60, ☐ 60 and older

☐ Single ☐ Married ☐ Children

How do you plan to use your spa/bath?

[A] Recreation [B] Entertainment

[C] Health Therapy [D] Business/Commercial

How did you find out about Pacific Marquis?

[A] Friend [B] Newspaper [C] Magazine [D] TV

[E] Radio [F] Other _____

Have you owned a pool or spa before? YES NO

How do you plan to install your spa?

[A] Self [B] Contractor [C] Dealer

MAIL PROMPTLY:

MARQUIS CORP.
596 HOFFMAN RD.
INDEPENDENCE, OR 97351
(503) 838-0888